

ADDRESSABLE STORAGE WITHOUT USABILITY: A PERMUTATION- RECURRENCE ESN DECODES DEEPER AND PREDICTS WORSE

A P R E P R I N T

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4 seeds, 12/12 strict inequalities

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ABSTRACT

We compare a reservoir designed for addressed storage — an echo state network whose recurrence is a single random permutation, $W = \rho \cdot P$, with Hadamard-coded inputs — against a sparse random-mixing echo state network on identical text, identical readouts and identical protocol. The addressed-memory network stores history **strictly deeper** (its past is linearly decodable further back) and predicts the next character **strictly worse**, in all four seeds tested, at both spectral radii, 12/12 strict inequalities on replication. The gap grows with spectral radius: depth $+0.9 \rightarrow +3.0$ characters, prediction cost $+0.07 \rightarrow +0.15$ bits. A follow-up shows the failure is structural: supplying the readout with explicit linear unbinding features for every addressed slot is a readout reparametrization and cannot help even in principle — the gap survives ($0.081 \rightarrow 0.091$ at $\rho=0.95$). A crossover sweep locates where the trade inverts: below $\rho^* \approx 0.72$ the pure delay line is the more usable arm. Storing information and using it are separable properties, dissociated here by construction.

Keywords reservoir computing · echo state networks · permutation recurrence · holographic reduced representations · decodability vs. usability · pre-registration

§1 Introduction

The program's reference reservoir is an echo state network with $N = 1024$ fixed random tanh units and sparse random recurrence, read over text one character at a time — a reservoir computer in the classical echo-state sense [1][2][3]. The recurrent weights are never trained; only a small readout on top of the state is. Such a reservoir *mixes*: every step distributes the incoming character across a shared, entangled state.

An alternative is to make the recurrence address its storage rather than mix it. Replace the mixing matrix with $W = \rho \cdot P$, where P is a single random 1024-cycle permutation: the state no longer mixes but *rotates*, each step advancing the whole stored history one addressed slot along the cycle. The inputs are given Hadamard codes — 27 exactly orthogonal input columns (scaled $1/\sqrt{3}$), one per character — so that successive writes do not interfere: the permutation-and-superposition storage scheme of holographic reduced representations [4] and hyperdimensional computing [5], built into the recurrence

itself. We refer to this construction as the **addressed-memory ESN**: the character seen j steps ago occupies slot j of the cycle, undisturbed. Whether such organized storage helps a trained readout is the question this study tests.

Two instruments are used, and the finding lives in their disagreement:

- **Decode depth** (U3 in the program's internal instrument numbering): how many characters back a ridge probe can still recover from the state — the interpolated 50%-accuracy crossing over lags 1–32.
- **Usability**: validation bits per character (bpc) of the standard logistic readout predicting the *next* character. Lower is better. For scale on these splits: uniform guessing 4.7549, 3-gram 2.9279, 5-gram 2.1585.

§2 Method

The study is a single registered $2 \times 2 (\times 2)$ factorial of 24 cells: recurrence $W \in \{\text{perm: } \rho \cdot P; \text{ sparse: fanin-10 random, rescaled to spectral radius } \rho\} \times \text{input coding } W_{\text{in}} \in \{\text{had: Hadamard; rnd: uniform random}\} \times \text{activation } \{\text{tanh, linear}\}$, with tanh evaluated at $\rho \in \{0.6, 0.95, 1.1\}$. Data: 2M training / 500k validation / 500k test characters of text8 [6], washout 200. In the short labels used in the tables below, `perm_had` denotes the addressed-memory ESN (permutation recurrence with Hadamard input coding) and `sparse_rnd` the sparse random-mixing control. One structural property is worth stating: a permutation is norm-preserving, so the addressed-memory arm remains stable at $\rho > 1$, beyond the spectral radius at which mixing reservoirs normally lose the echo state property.

§3 Pre-registered predictions

Six ordinal predictions and four quantitative bands were committed before the run, with wording finalized after a disclosed 100k-character pilot run — a disclosure the registration protocol requires. The headline prediction, O6: the addressed-memory ESN would be *both* strictly deeper on decode depth *and* strictly worse on logistic bpc than the sparse-mixing reservoir at $\rho \in \{0.95, 1.1\}$ — the claim that addressable storage is bought at the cost of mixing, stated as a conjunction so that either half failing would break it.

Because O6 was a surprising claim, the program's standing rules held it provisional until a registered replication on seeds 1–3: 6 cell-pairs, 12 strict inequalities, with the falsifiers named in writing before the rerun — *any* depth inequality breaking would demote the claim to an open question; any bpc inequality breaking with the depth ordering intact would half-falsify it; both breaking would record seed 0 as a candidate outlier. 12/12 was the only outcome permitted to promote the finding to the program's confirmed-findings record.

§4 Results

Run #1 (seed 0): 6/6 ordinal predictions confirmed, 4/4 quantitative bands confirmed. The registered headline, in numbers:

Table 1. Addressed-memory ESN (perm_had, tanh) vs sparse random-mixing ESN (sparse_rnd, tanh), seed 0. Depth in characters (higher = decodes deeper); logistic validation bpc (lower = predicts better).

ρ	depth perm_had	depth sparse_rnd	bpc perm_had	bpc sparse_rnd
0.95	11.05	10.11	2.7538	2.6727
1.1	13.72	10.63	2.8779	2.7337

Deeper in the left-hand columns, worse in the right-hand columns, at both radii — and the depth gap along the W axis *grows* with ρ (roughly $+0.4 \rightarrow +1.0 \rightarrow +3.2$ characters across $\rho = 0.6 / 0.95 / 1.1$), with no collapse of the addressed arm near the stability boundary. The replication, seeds 1–3:

Table 2. Registered replication (internal run log #2), seeds 1–3. Every registered inequality strict: 12/12.

seed	ρ	depth: perm > sparse	bpc: perm > sparse
1	0.95	11.04 > 10.10	2.7613 > 2.6681
1	1.1	13.70 > 10.64	2.8833 > 2.7303
2	0.95	11.02 > 10.12	2.7480 > 2.6751
2	1.1	13.68 > 10.70	2.8699 > 2.7360
3	0.95	11.03 > 10.07	2.7565 > 2.6671
3	1.1	13.67 > 10.63	2.8744 > 2.7209

The seed-to-seed spread is notably small: across four seeds the addressed-memory ESN's depth spans 11.02–11.05 at $\rho=0.95$ and 13.67–13.72 at 1.1 — a spread of ≤ 0.05 characters, roughly half a percent of the gap under test. A plausible explanation is that a random N -cycle is the same object up to relabeling, so the decode depth of an addressed-memory ESN is essentially deterministic in the seed.

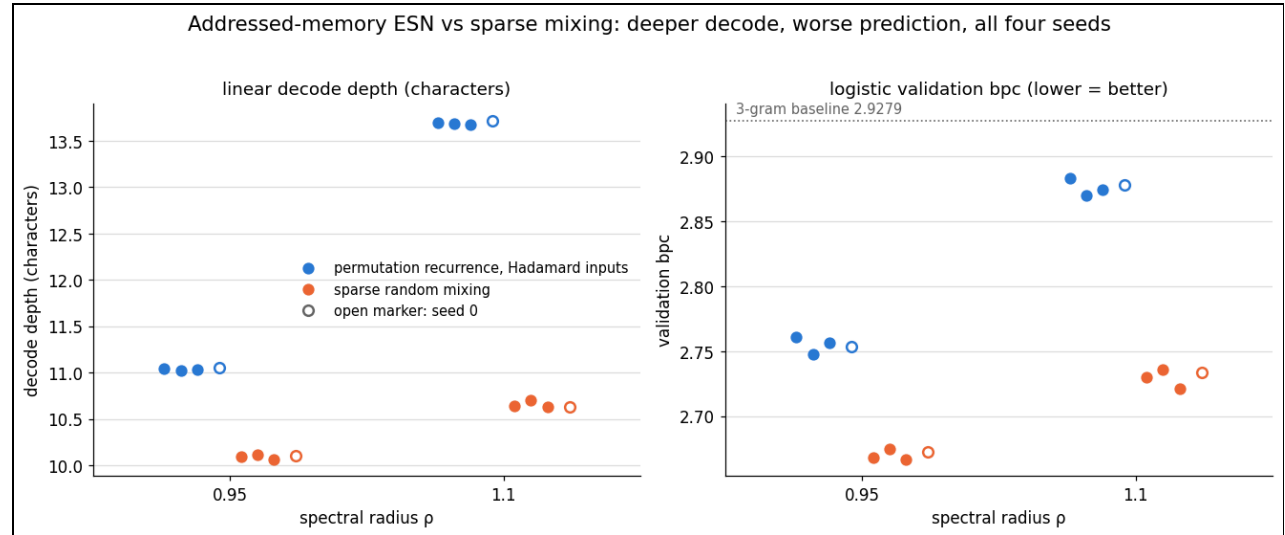


Figure 1. Summary of the replication (internal run log entries #1–#2, tanh arms). Left: U3 decode depth (characters) vs ρ . Right: logistic validation bpc vs ρ ; dotted line: the 3-gram baseline 2.9279. Blue: perm_had (addressed-memory ESN); orange: sparse_rnd (sparse random mixing); filled markers: seeds 1–3; hollow: seed 0 (run #1). At both ρ the addressed-memory ESN decodes deeper in the left panel and predicts worse in the right panel in every seed — the 12/12 strict inequalities of Table 2.

Can the extra depth be made usable by explicit unbinding? A natural objection is that the readout cannot address the stored slots directly. We therefore supplied the addresses explicitly — features $[x \parallel P^{-1}x \parallel \dots \parallel P^{-4}x]$, the state unbound in the holographic-representation sense [4] by the reservoir's own permutation cycle. The registered analysis established this to be vacuous before the run: each appended block is a fixed coordinate permutation of x , so the widened features span *exactly* the raw state's linear function class — a linear readout already realizes every fixed permutation of its input. The run confirmed the observable consequence: ridge validation bpc identical to four decimal places in every arm, and the permutation–sparse logistic gap *survived*, $0.0811 \rightarrow 0.0906$ at $\rho=0.95$ and $0.1443 \rightarrow 0.1484$ at $\rho=1.1$. Linear unbinding cannot recover the stored depth even in principle. (A nonlinear decoder is a named, not-yet-run follow-up.)

Where the trade inverts. A registered low- ρ sweep (seed 0, random input coding in both arms, isolating the W axis) measured the gap function $\text{gap}(\rho) = \text{sparse} - \text{perm logistic validation bpc}$ (positive = the permutation delay line is the more usable arm):

Table 3. The mixing-cost crossover (internal run log #6, seed 0; single-seed measured context).

ρ	perm_rnd bpc	sparse_rnd bpc	gap	winner
0.3	2.8969	2.9028	+0.0059	delay line
0.45	2.7758	2.7865	+0.0107	delay line
0.6	2.6843	2.6989	+0.0146	delay line
$\rho^* = 0.7227$	–	–	0	crossover
0.8	2.6611	2.6519	–0.0092	mixing
0.95	2.7353	2.6727	–0.0626	mixing

Below $\rho^* \approx 0.72$ the pure delay line is the *more usable* arm; above it, mixing wins and the cost of the permutation grows. The storage ordering — the addressed-memory ESN decodes deeper — held at every ρ tested, 0.3 through 1.1. Both architectures have their bpc optimum near $\rho=0.8$, just above the crossover: at the operating point where these reservoirs predict best, addressed storage and mixing are nearly equivalent, and they separate only away from it — mixing degrades gracefully, while the delay line collapses.

§5 Discussion

Storing information and using it are separable properties, and here the trade is imposed by construction: the permutation buys addressable, collision-free storage at the cost of nonlinear mixing, and the readout pays for the missing mixing in bits. The limitation is structural rather than a missing tool — supplying explicit unbinding features does not help, because applying a fixed permutation is an operation a linear readout already performs implicitly. What the prediction readout requires is not deep, linearly decodable storage but the computed, entangled summaries that mixing produces. Decodability is not usability.

One qualification remains open: only *linear* unbinding is ruled out. Whether an inexpensive *nonlinear* decoder can convert the stored depth into predictive performance is a named open question in the program's queue, and no claim is made about it here.

§6 References

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 2. W. Maass, T. Natschläger, H. Markram, "Real-time computing without stable states: a new framework for neural computation based on perturbations," *Neural Computation* 14(11), 2002.
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 5. P. Kanerva, "Hyperdimensional computing: an introduction to computing in distributed representation with high-dimensional random vectors," *Cognitive Computation* 1(2), 2009.
 6. M. Mahoney, "Large text compression benchmark" (text8), matmahoney.net/dc/textdata.
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§7 Provenance

- **169b7c2d** — 2026-08-10-addressed-pond: main factorial, 24 cells, seed 0 (internal run log #1). Registered before running; all six ordinal predictions and four quantitative bands confirmed.
- **866d975e** — 2026-08-11-addressed-seeds-replication: seeds 1-3, 12 cells (internal run log #2). 12/12 strict inequalities; promoted by the pre-stated rule.
- **472d4fb7** — 2026-08-11-addressed-unbinding-readout: the linear unbinding test, seed 0 (internal run log #5). Null registered in advance and confirmed; this line of inquiry closed.
- **4052cdec** — 2026-08-11-addressed-low-rho-crossover: the crossover sweep, seed 0 (internal run log #6). Single-seed measured context, labeled as such.

All experiments registered before running, with named falsifiers. Headline claim tested on 4 seeds (0-3); replicated before publication. Unbinding and crossover results are single-seed and presented as supporting structure, not as the promoted claim. Internal designation: addressed pond.

AuxiLab · findings are pre-registered and replicated before publication

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